الجمهورية الجزائرية الديمقراطية الشعبية

People's Democratic Republic of Algeria

Ministry of Higher Education and Scientific Research
National School of Cybersecurity

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NATIONAL SCHOOL OF CYBERSECURITY

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قسم التكوين القاعدي

Technical Report – System Audit script

Course: Operating Systems (OS2)
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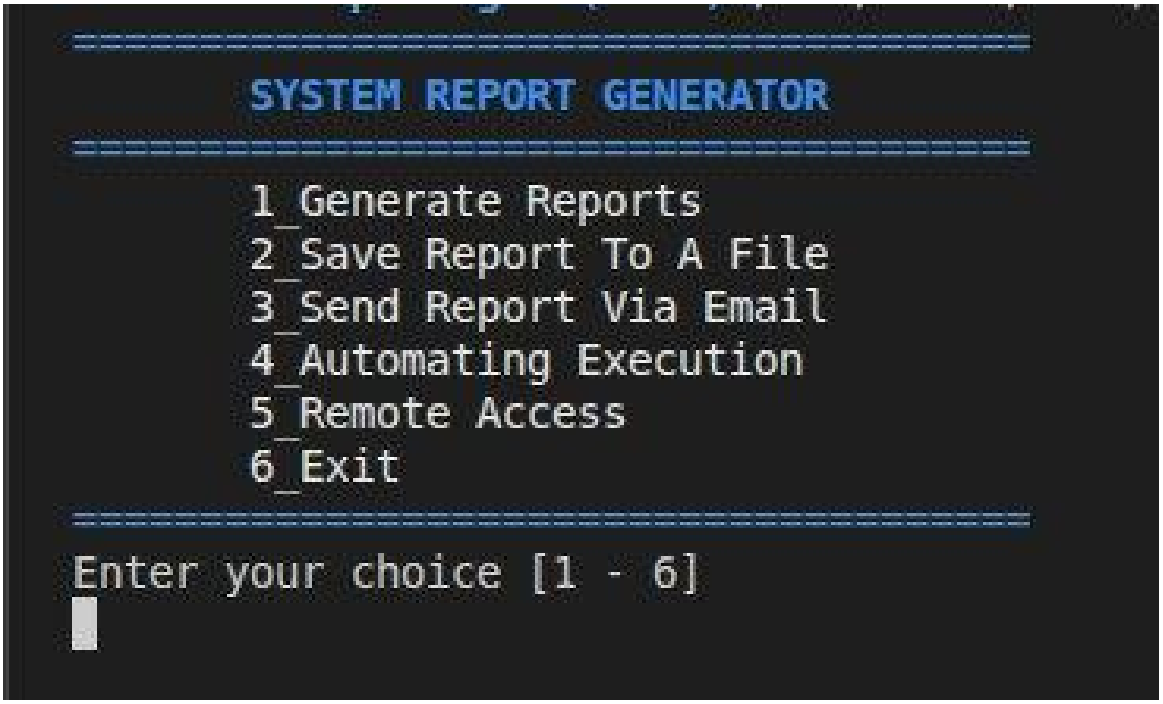
Date: April 2026



1. Introduction:

This report describes a Linux-based project that automatically collects hardware and software information from a computer, generates organised reports, and provides features such as email sending, scheduled execution, and remote monitoring. The work was carried out as part of the Operating Systems course at the National School of Cybersecurity.
The main goal of the project is to help system administrators and cybersecurity professionals keep track of what is installed on a machine, detect unexpected changes, and produce documentation that can be used for incident response or asset management. Instead of running commands manually every time, this tool automates the whole process and makes the output readable and well structured.

2. Design Explanation:

The project is divided into several folders and files to keep things organised. The main entry point is `main.sh`, which shows a menu and lets the user choose what to do. All the actual work is done by functions stored in separate files inside the `FUNCTION_FOLDER`. Report generation scripts are kept in the `REPORTS_FOLDER`, so it is easy to see which script does what.

The main menu offers six options:

1. Generate reports (hardware or software, short or detailed).
2. Save a copy of an existing report to a location chosen by the user.
3. Send a report by email, with the possibility of attaching it as a PDF.
4. Schedule automatic report generation using cron.
5. Connect to a remote machine via SSH and run the audit there.
6. Exit the programme.

When the user selects the first option, a submenu appears where they can choose between hardware and software reports, and then between a short summary or a full detailed version. After that, the appropriate report script is executed. The output is immediately shown on the screen and also saved to a file with a timestamp so that old reports are never overwritten.

2.2 Hardware Reports:

The hardware part collects information about the main components of the machine:

- CPU: model, number of cores, architecture, current speed, maximum and minimum speed, flags that indicate supported features, security vulnerabilities, and cache details (L1, L2, L3).

- Memory: total RAM, used and available memory, swap space, memory type (for example DDR4), speed, number of physical slots, how many are occupied, the manufacturer, form factor, and maximum supported capacity.

- Storage:disk model, total size, whether it is an SSD or an HDD, interface type (SATA, NVMe, USB), serial number, partitions, filesystem types, mount points, and UUIDs.

- Network: interface names, IPv4 addresses, MAC addresses, network card model, driver, and link speed.

- USB devices:controllers, device tree, speeds, serial numbers, manufacturer and product names, and total device count.

- Motherboard:manufacturer, product name, serial number, version, chipset, southbridge, SATA controller, and audio controller.

- BIOS/UEFI: vendor, version, release date, boot mode (UEFI or legacy), Secure Boot status, TPM presence, and SMBIOS version.

- System overview:kernel version, system uptime, last boot time, virtualization type, and chassis type.

- Graphics: GPU vendor, model, driver, video memory size, number of GPUs, and temperature if sensors are available.

The short hardware report contains only the most essential information (model, total memory, disk usage, etc.), while the detailed version adds all the extra fields.

2.3 Software Reports:

The software part focuses on the operating system and what is running on it:

- Current logged-in users and a complete list of system users.
- Running processes (top five in the short report, all of them in the detailed report).
- Active system services.
- Open network ports.
- Number of installed packages

Like the hardware reports, the short version gives a quick overview and the detailed version shows everything.

2.4 Email Sending:

When the user chooses option 3, the script asks for the recipient's email address, a subject line, and an optional message. The user then selects which report to send. The script looks for the most recent report of that type, removes the colour codes that are used for terminal display, converts the cleaned text into a PDF using the `enscript` and `ps2pdf` commands, and finally sends the email with the PDF attached. The email is sent with `mutt` in batch mode so that no interactive screen appears, and `msmtp` is used as the actual SMTP client.

2.5 Cron Automation:

Option 4 lets the user schedule reports to run automatically. The user chooses which report and enters the hour and minute when it should be executed. The script then creates a small wrapper script that contains the exact command to run, and adds it to the user's crontab. The user can also view existing cron jobs or remove them all.

2.6 Remote Monitoring:

Option 5 is designed to work with remote machines. The user can add information about a remote computer (hostname, username, SSH port, and the path to the script on that machine). Once a machine is added, it is possible to run the audit manually on that machine or to schedule it automatically via cron. When the remote report finishes, the generated file is copied back to the local machine using `scp`.

3. Commands Used:

3.1 Software Information:

OS | `cat /etc/os-release` --> OS name and version .
Kernel | `uname -r` --> kernel version .
Users | `who` and `cut -d: -f1 /etc/passwd` --> logged-in users and all system users
Processes | `ps aux` --> list of running processes
Services | `systemctl list-units --type=service` --> active services .
Ports | `ss -tuln` --> open network ports .
Packages | `dpkg -l` --> installed packages .

3.2 Hardware Information:

CPU | `lscpu` --> model, cores, architecture, sockets, cache
CPU | `cat /proc/cpuinfo` --> current speed, flags, vulnerabilities
Memory | `free -h` --> total, used, available, swap
Memory | `sudo dmidecode -t memory` --> type, speed, slots, manufacturer
Storage | `lsblk` --> disk model, size, partitions, interface Storage | `df -hT` --> usage, filesystem type, mount points
GPU | `lspci \| grep -E "VGA\|3D\|Display"` --> model and vendor
Network | `ethtool` --> link speed and duplex
USB | `lsusb` and `lsusb -t` -->connected devices and device tree
Motherboard | `sudo dmidecode -s baseboard-*` --> manufacturer, product, serial
BIOS | `sudo dmidecode -s bios-*` --> vendor, version, release date
System | `uname -r`, `uptime -p`, `who -b` --> kernel, uptime, last boot

3.3 Email and PDF Tools:

msmtp : sends email using an SMTP server
mutt : creates and sends emails, handles attachments
enscrip` : converts text to PostScript
ps2pdf: converts PostScript to PDF

3.4 Automation and Remote Access:

| `crontab -e`, `crontab -l`, `crontab -r` | manage scheduled jobs |
| `ssh` | connect to a remote machine |
| `scp` | copy files between machines |
| `ssh-keygen`, `ssh-copy-id` | set up passwordless SSH access |

4. Cron Configuration and Screenshot:

The cron job is set to run the hardware short report every day at 4:00 AM. The line added to the crontab is:
/home/vboxuser5/ASDworkspace/REPORTS_FOLDER/HARDWARE_INFO_SHORT.sh

Because cron uses a minimal environment, a few extra settings were necessary. The following lines were added at the top of the crontab file:

PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
SUDO_ASKPASS=/home/vboxuser5/askpass.sh

The `PATH` line ensures that commands like `lscpu` and `lsblk` are found. The `SUDO_ASKPASS` line makes it possible for `sudo` to work inside the script without asking for a password.

```
57 17 * * * /home/vboxuser5/ASDworkspace/REPORTS_FOLDER/HARDWARE_INFO.sh
57 17 * * * /home/vboxuser5/ASDworkspace/REPORTS_FOLDER/HARDWARE_INFO_SHORT.sh
59 17 * * * /home/vboxuser5/ASDworkspace/REPORTS_FOLDER/HARDWARE_INFO.sh
58 15 * * * /home/vboxuser5/ASDworkspace/REPORTS_FOLDER/SOFTWARE_INFO.sh
0 18 * * * /home/vboxuser5/ASDworkspace/remote_ubuntu2.sh
03 18 * * * /home/vboxuser5/ASDworkspace/REPORTS_FOLDER/SOFTWARE_INFO_SHORT.sh
04 18 * * * /home/vboxuser5/ASDworkspace/REPORTS_FOLDER/SOFTWARE_INFO.sh
05 18 * * * /home/vboxuser5/ASDworkspace/remote_ubuntu2.sh
25 18 * * * /home/vboxuser5/ASDworkspace/remote_ubuntu2.sh
59 20 * * * /home/vboxuser5/ASDworkspace/REPORTS_FOLDER/HARDWARE_INFO_SHORT.sh
```

5.Security Considerations:

The email configuration is stored in `~/.msmtprc`. This file contains the Gmail address and an App Password that was generated specifically for this project. The file is protected with `chmod 600` so that only the owner can read it. No password is written directly inside any of the scripts.

Instead of running the whole script as root, cron uses the normal user account, but some commands (like `dmidecode`) require extra privileges. To avoid being asked for a password every time, a specific line was added to `/etc/sudoers` using `visudo`:vboxuser5 ALL=(ALL) NOPASSWD: /usr/sbin/dmidecode, /bin/mkdir, /usr/bin/tee

This follows the principle of least privilege: only the necessary commands are allowed to run without a password, not every possible command.

6.Challenges Encountered and Solutions:

6.1 Cron Would Not Run the Script:

At first, the cron job was installed but the script never produced any output. The reason was that cron uses a very limited `PATH` and does not know where to find commands like `lscpu` or `lsblk`. Adding a `PATH` line at the top of the crontab fixed the problem. Also, the script itself was modified to change to its own directory (`cd /home/vboxuser5/ASDworkspace`) so that relative paths to the function files would work.

6.2 `sudo` Inside Cron Asked for a Password:

When the script tried to use `sudo` (for `dmidecode`), the cron environment did not have a terminal and could not ask for a password. The error message clearly said that a terminal was required. The solution was to create a small `askpass.sh` script that echos the password, set the `SUDO_ASKPASS` variable in crontab, and change all `sudo` commands to `sudo -A`. This tells `sudo` to use the askpass helper.

6.3 Remote Monitoring Kept Asking for a Password

Even though SSH from the local machine to the remote machine worked without a password, the `scp` command that copied the report back was still asking for a password. The reason was that the remote machine did not have a key to authenticate back to the local machine. Installing a second SSH key on the remote side and copying it to the local machine solved the problem.

6.4 Hardware Information Missing in the VM

Some hardware details, such as the exact memory type or the number of memory slots, were not available because the system runs inside a VMware virtual machine. The hypervisor hides some of the physical hardware information. Instead of leaving those fields blank, the script now explicitly prints a message saying that the information is not available in a virtualised environment.

7.Conclusion:

The System Audit Script fulfills all the requirements of the OS2 mini-project. It collects a wide range of hardware and software information, presents it in a clean and readable format, and provides several useful features such as email sending, automatic scheduling, and remote monitoring. The project is organised in a modular way, which makes it easy to maintain and extend.

During development, several interesting problems were encountered, especially related to cron, sudo, and email automation. Each of these problems was analysed and solved using standard Linux tools and techniques. The final script works reliably in a Linux environment, whether run manually from the terminal or automatically through cron.

The project demonstrates a good understanding of shell scripting, Linux system administration, and security practices such as least privilege, proper file permissions, and the use of App Passwords instead of storing real passwords in code.

The code is available on GitHub, and this report together with the example outputs and the technical documentation makes the project ready for submission.